Nathan Nguyen

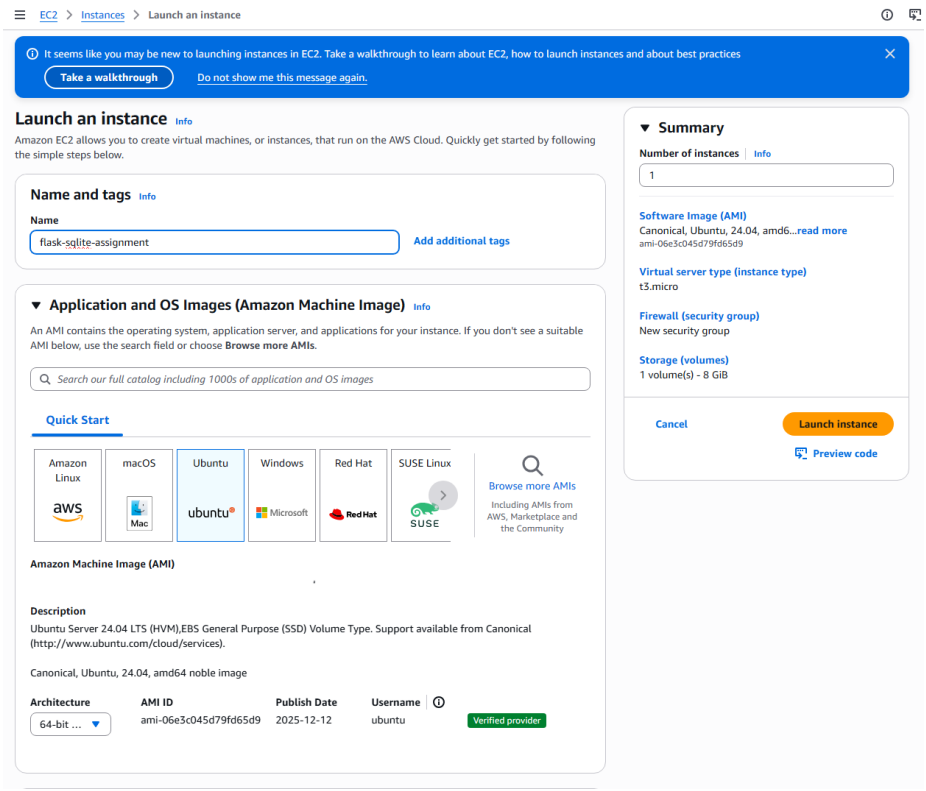
AWS Project Submission

1. **Launch an EC2 instance (any platform accepted, but it should be accessible through the internet). (Points: 1)**
    - (Screenshots of each major step: AWS portal setups, EC2 creation, Amazon Machine Image (AMI), key pair (login), network settings, etc.)
2. **Configure a web server and SQLite3 database table (any variation, but help is available for limited variations). (Points: 1)**
    - (Screenshots of Flask application setup in the EC2 instance, installation of mod_wsgi for Apache, installation of Python, Pip, and Flask, SQLite3 database table, etc.)

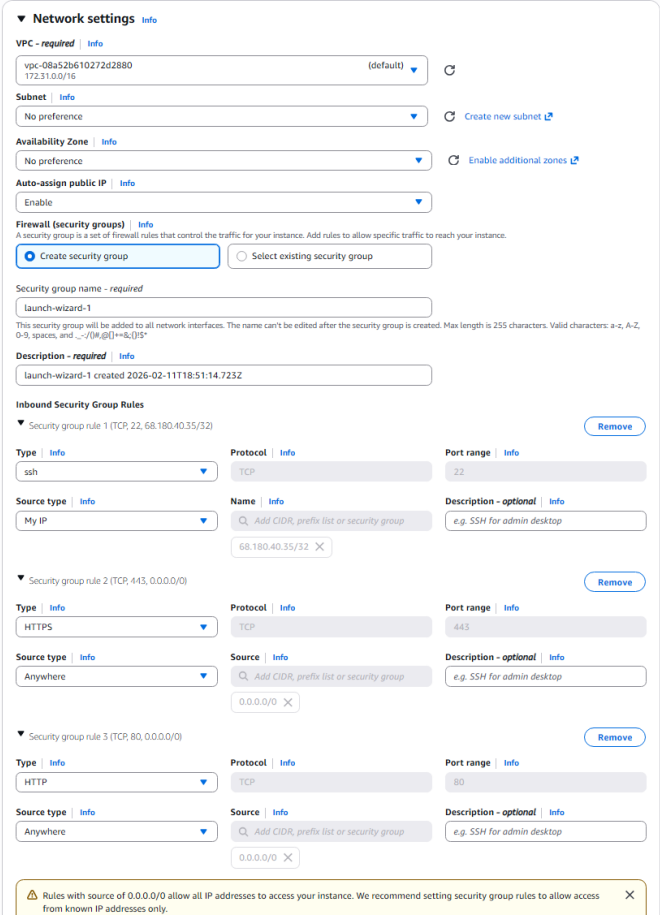
Photos of key Steps for 1-2:

Step 1:

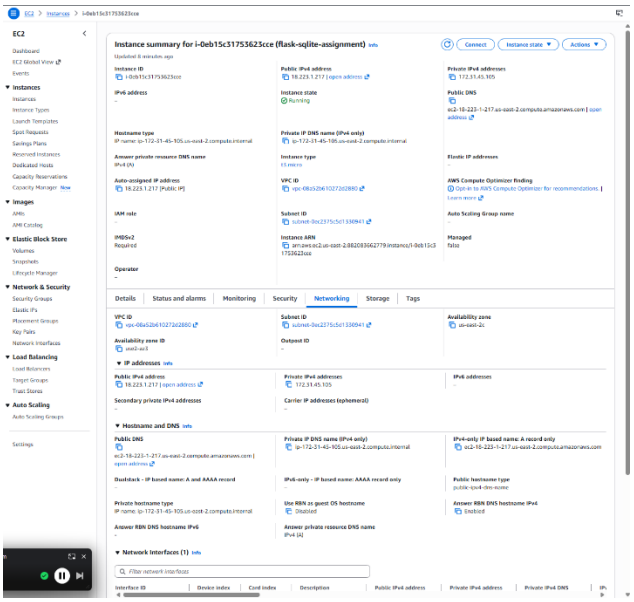
Creating EC2 Instance



EC2 instance network settings



EC2 & AMI creation



Key pair login test success

```
→ Downloads ssh -i ~/.ssh/project2-aws.pem ubuntu@ec2-18-223-1-217.us-east-2.compute.amazonaws.com
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Feb 11 19:11:59 UTC 2026

  System load:  0.0                Temperature:           -273.1 C
  Usage of /:   12.1% of 14.46GB   Processes:             111
  Memory usage: 23%                Users logged in:       0
  Swap usage:   0%                 IPv4 address for ens5: 172.31.45.105

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-45-105:~$
```

Step2:

Installation setup for EC2 instance

```
ubuntu@ip-172-31-45-105:~$ sudo apt-get update -y
sudo apt-get install -y apache2
sudo apt-get install -y libapache2-mod-wsgi-py3
sudo apt-get install -y python3-pip python3-flask
sudo apt-get install -y sqlite3
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1741 kB]
Get:14 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [324 kB]
Get:15 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:16 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.5 kB]
Get:17 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1528 kB]
Get:18 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [313 kB]
Get:19 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.9 kB]
```

Checking if installed correctly

```
ubuntu@ip-172-31-45-105:~$ apache2 -v
python3 --version
sqlite3 --version
Server version: Apache/2.4.58 (Ubuntu)
Server built:   2025-12-09T15:50:28
Python 3.12.3
3.45.1 2024-01-30 16:01:20 e876e51a0ed5c5b3126f52e532044363a014bc594cfefa87ffb5b82257ccalt1 (64-bit)
```

SQLLite3 database table setup + flask app

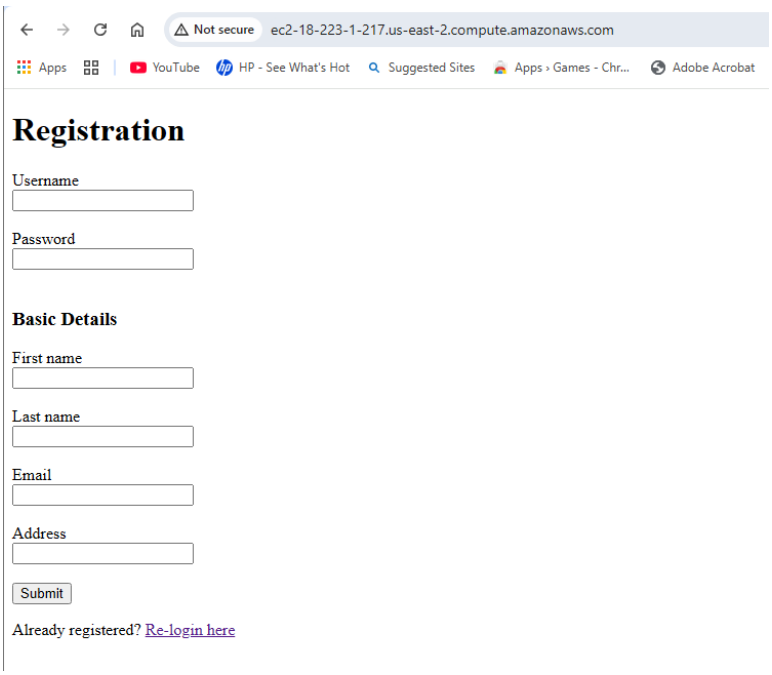
```
  GNU nano 7.2                    /home/ubuntu/flaskapp/flaskapp.py
import os
import sqlite3
from flask import Flask, render_template, request, redirect, url_for, send_from_directory, flash

APP_DIR = os.path.dirname(os.path.abspath(__file__))
DB_PATH = os.path.join(APP_DIR, "users.db")

UPLOAD_DIR = os.path.join(APP_DIR, "static", "uploads")
DOWNLOAD_DIR = os.path.join(APP_DIR, "static", "downloads")

app = Flask(__name__)
app.secret_key = "replace-this-with-any-random-string"

def init_db():
    conn = sqlite3.connect(DB_PATH)
    c = conn.cursor()
    c.execute("""
        CREATE TABLE IF NOT EXISTS users (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            username TEXT UNIQUE NOT NULL,
            password TEXT NOT NULL,
            firstname TEXT NOT NULL,
            lastname TEXT NOT NULL,
            email TEXT NOT NULL,
            address TEXT NOT NULL,
            uploaded_filename TEXT,
            uploaded_word_count INTEGER
        )
    """)
    conn.commit()
    conn.close()

def get_user_by_username(username: str):
    conn = sqlite3.connect(DB_PATH)
    c = conn.cursor()
    c.execute("""
        SELECT username, password, firstname, lastname, email, address, uploaded_filename, uploaded_word_count
        FROM users
        WHERE username = ?
```

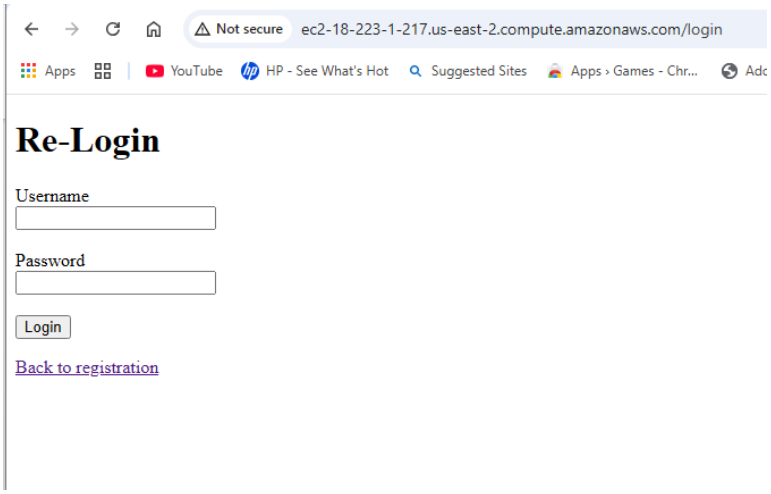
Check for table is correct

```
sqlite> SELECT username, firstname, lastname, email, address, uploaded_word_count FROM users;
nathanfat|Nathan|Nguyen|Nathanfire100@gmail.com|1854 Knollmont Dr|32
```

4. **Share the code on GitHub or as an attachment.** (Example: https://github.com/UserNameLinks to an external site.) (Points: 1)

    - Link to code: https://github.com/Nathann03/CS5165-Cloud_computing/tree/main/project2/code
    - Link to AWS URL: http://ec2-18-223-1-217.us-east-2.compute.amazonaws.com/





Not secure ec2-18-223-1-217.us-east-2.compute.amazonaws.com/profile/nathanfat

Apps YouTube HP - See What's Hot Suggested Sites Apps › Games - Chr... Adobe Acrobat

User Information

Username: nathanfat

First name: Nathan

Last name: Nguyen

Email: Nathanfire100@gmail.com

Address: 1854 Knollmont Dr

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Choose File No file chosen Upload

Stored file: nathanfat_Limerick.txt

Word count: 32

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